

case demo

research showcase (?) + how to use AI for ur app :)

Tao 🤠



- 3rd year CS PhD working on **HCI** w/ **Lydia Chilton**
- Research on human-AI interaction design & creativity support tools
- TA-ed UI design, Designing for GenAI, Advanced Web Design Studio
- From Shenzhen, China, Undergrad @ Cornell
 - information science & communication
 - graduated in 2022, so i'm not that old lol, i also listen to olivia rodrigo & many kpop hehe
- Talk to me about...
 - anything :)
 - NYC concerts (i went to 5 concerts last summer and helped hold one in central park)
 - food (especially the all you can eat ones)
 - free trips (i went for free trips almost every 2 months)
 - PhD (maybe!)

Human-Computer Interaction (HCI)

Human–Computer Interaction (HCI)

- Cross-disciplinary field that studies how computers should be designed

Human-Computer Interaction (HCI)

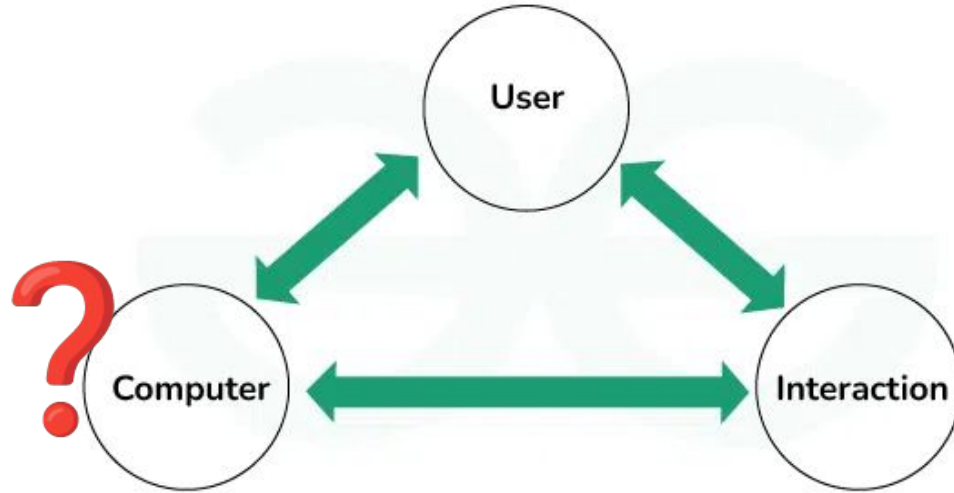
Human–Computer Interaction (HCI)

- **Cross-disciplinary field that studies how computers should be designed**
 - What computers are used for
 - How people interact with computers
 - How either of these two things should change in the future

Human-Computer Interaction (HCI)

Human-Computer Interaction (HCI)

- Cross-disciplinary field that studies how computers should be designed



<https://www.geeksforgeeks.org/introduction-to-human-computer-interface-hci/>

Definition of “Computer” Over the Years

Definition of “Computer” Over the Years

- **Pre-1940s:** People performing calculations



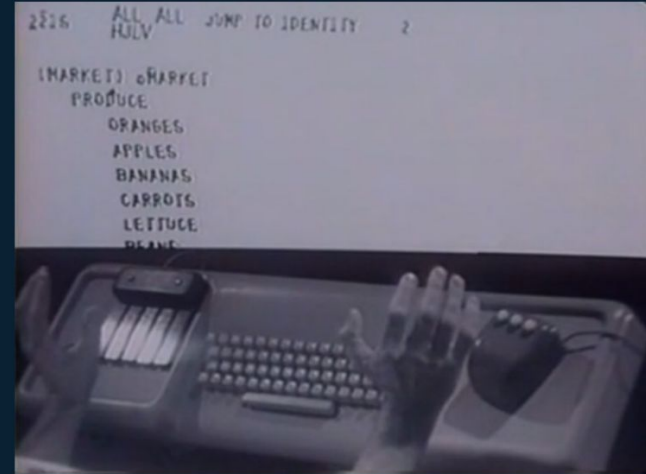
NACA Computers (1949)

https://upload.wikimedia.org/wikipedia/commons/0/06/Human_computers_-_Dryden.jpg

Slide adapted from Prof. Lydia Chilton

Definition of “Computer” Over the Years

- **Pre-1940s:** People performing calculations
- **1940s–1960s:** Machines performing symbolic manipulations
- **1945:** Tools to augment human cognition
- **1968:** Interactive systems augmenting people’s abilities



Douglas Engelbart's
“Mother of All Demos” (1968)

<https://www.extremetech.com/wp-content/uploads/2013/07/douglas-engelbart-the-mother-of-all-demos-mouse-640x478.jpg>

Slide adapted from Prof. Lydia Chilton

Definition of “Computer” Over the Years

- **1968:** Interactive systems augmenting people’s abilities
- **Today:** Ubiquitous, wearable augmentations of people’s abilities



Focals by North
(2018)

Images: Focals, *The Verge*

Jacquard by Google (2018)

Image: <https://www.levistrauss.com/2018/01/18/story-behind-levis-jacquard-jacket/>

Definition of “Computer” Over the Years

- **1968:** Interactive systems augmenting people’s abilities
- **Today:** Ubiquitous, wearable augmentations of people’s abilities
- **Tomorrow: ?**

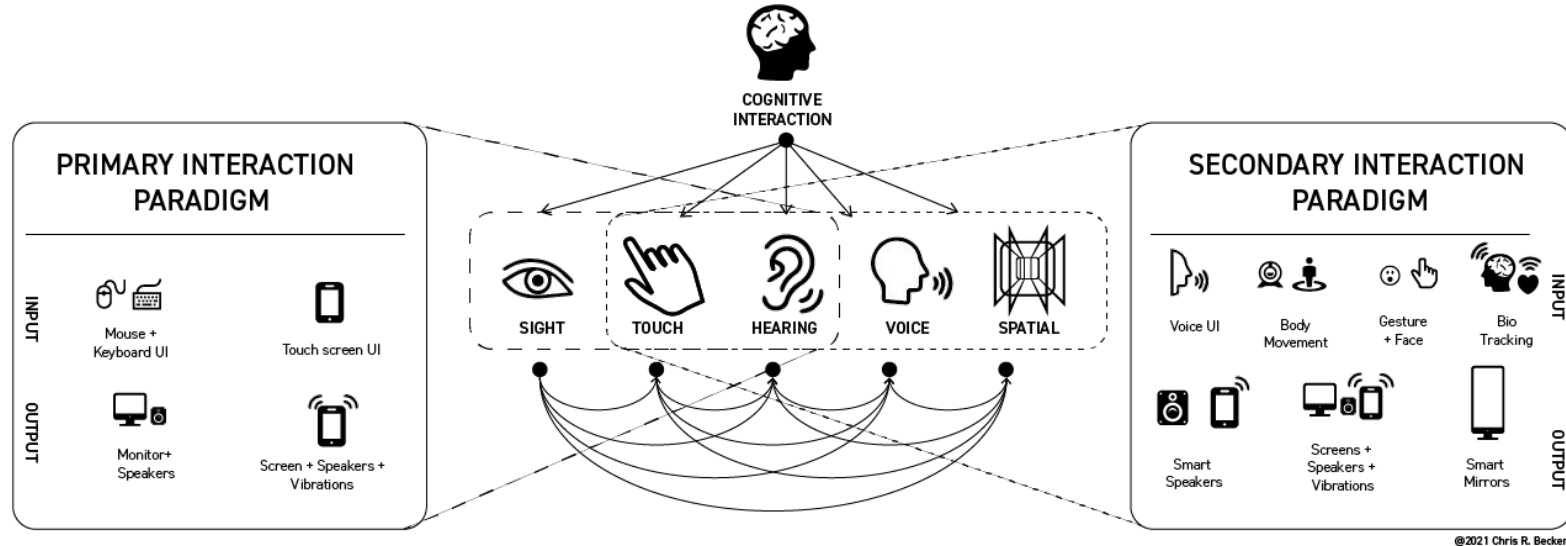


**HCI research has changed how we view computers,
even before we formalized it as a discipline!**

Slide adapted from Prof. Lydia Chilton

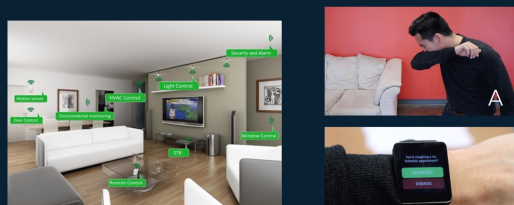
Human-Computer Interaction (HCI)

HUMAN INTERACTION WITH A COMPUTER

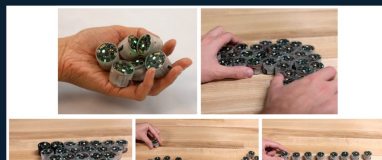


Human-Computer Interaction (HCI)

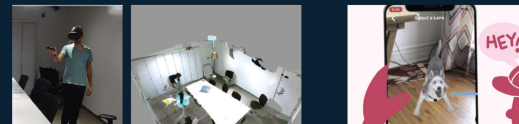
2. Ubiquitous Computing and Mobile Health



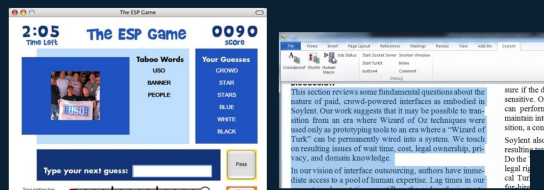
4. Tangible Computing and New Interaction Styles



5. Mixed Reality and Wearable Computing



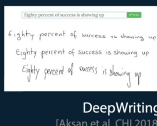
6. Collective Intelligence



11. Computational Interaction and Generative AI



Generative Agents: Interactive Simulacra of Human Behavior
[Park et al. UIST 2023]



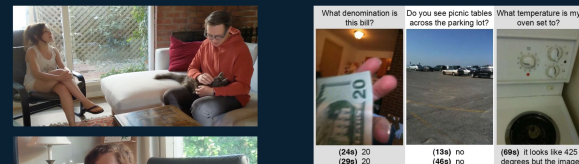
DeepWriting
[Aksan et al. CHI 2018]

12. Sensemaking and Creativity Support

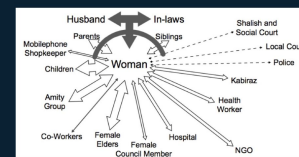


The Elements of Fashion Style
[Vaccaro et al. UIST 2016]

8. Accessibility



13. HCI and Society



Design Within a Patriarchal Society:
Opportunities and Challenges in
Designing for Rural Women in
Bangladesh
[Sultana et al. CHI 2018]



"A Stalker's Paradise": How Intimate
Partner Abusers Exploit Technology
[Freed et al. CHI 2018]

My research: creativity support and human-AI interaction

- How do creative professionals use AI?
 - (ie artists, writers, journalists(?), content creators in general)
- What are their experiences, challenges, and perception?

***I build creativity support tools to
assist creative professionals***

writing tool, design tool, etc.

My research: creativity support and human-AI interaction

AUDIO AREA

▶ Play

+

ADD INTERVAL

🗑

DELETE

💾

SAVE

dancing at the disco

🔍 BRAINSTORM

INTERVAL # 1

BEGIN TIME: 0.7

END TIME: 1.3

neon dance floor: Robot D.J.s, glitch

540

Robot D.J.s, glitch., neon dance floor

540

BRAINSTORMING AREA

SUBJECTS

Robot D.J.s, neon dance floor

Groovy dancers, colorful disco ball

Colorful lights, groovy dance moves

CHOOSE STYLES

kaleidoscope

motion blur

lens flare

cinematic lighting

front view

illustration

magazine cover

golden hour

slow motion

blue hour

sunrise

glitch

vaporwave

pattern

sketch

Robot D.J.s, neon dance floor, glitch, kaleidoscope

Seed (optional)

🔍 PREVIEW IMG

HISTORY

14_robot ballroom dancer

13_robot ballroom dancer

12_robot ballroom dancer

8_hip hop dancer, glitch, vaporwave, 878

3_glitch, Robot, D.J.s, ballroom, 440

2_Robot D.J.s, glitch, vaporwave, 440

2_dancing robot, 440

1_dancing robot, 440

0_dancing robot, 440

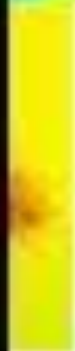
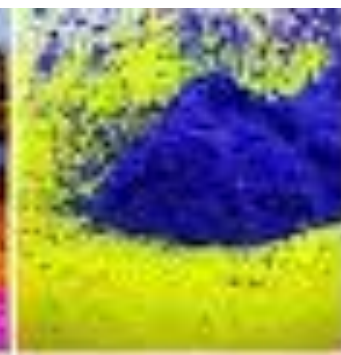
13_kaleidoscope, 440

1_neon dance

- music visualization
- start & end prompt
- transition, hold, jump cut

EMPIRE STATE OF MIND - Alicia Keys

[illegible]

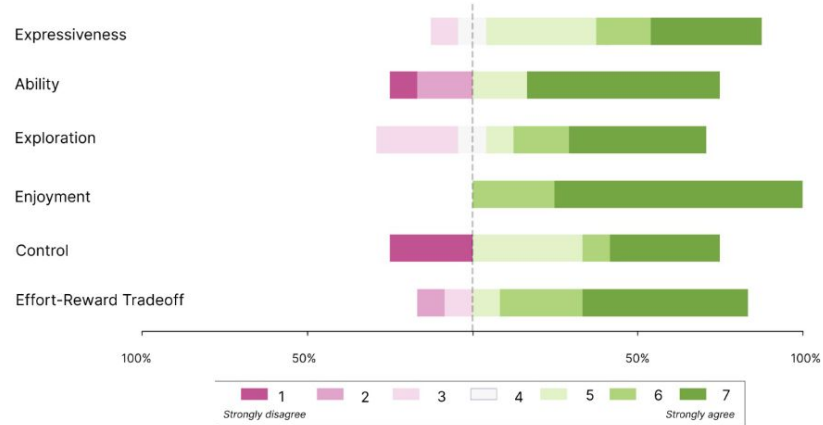




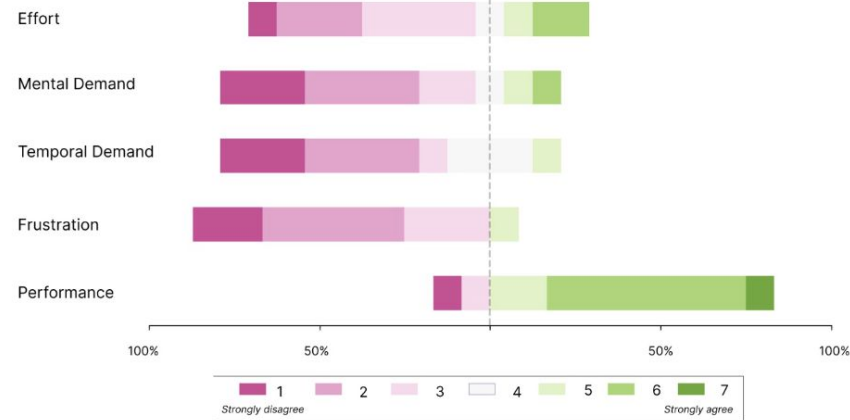


Creativity Support Metrics Across Music Visualization Task

How much do you agree or disagree with the following statement?

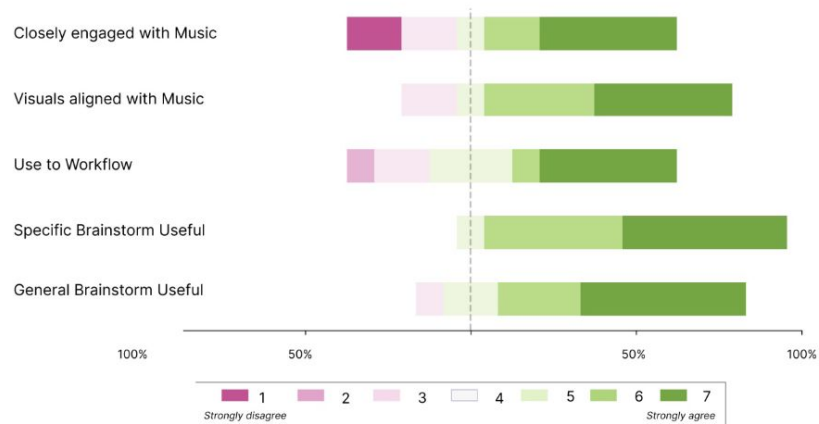


NASA -TLX Metrics Across Music Visualization Task



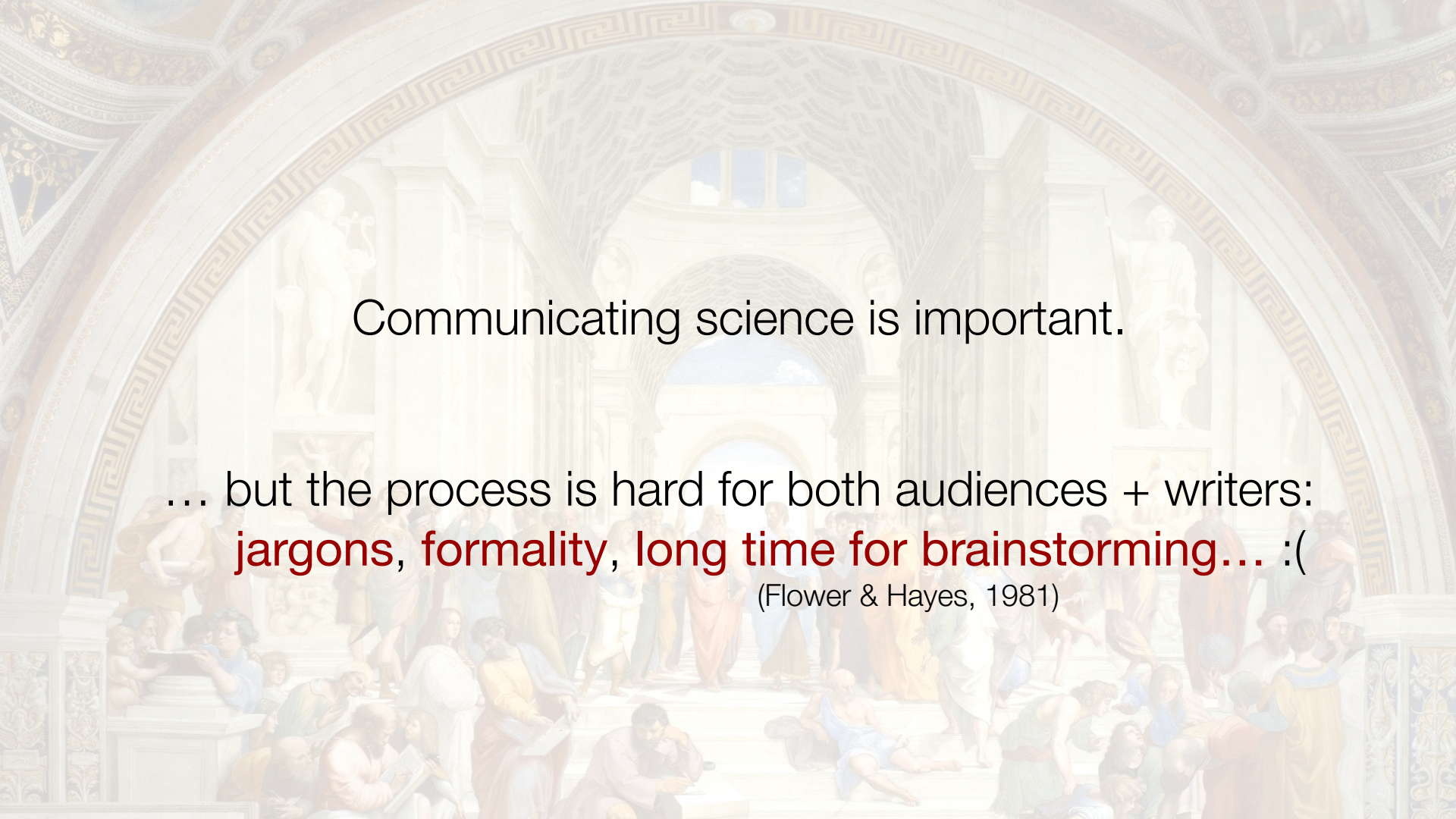
Workflow-Specific Questions

How much do you agree or disagree with the following statement?



science writing tool





Communicating science is important.

... but the process is hard for both audiences + writers:
jargons, formality, long time for brainstorming... :(

(Flower & Hayes, 1981)

now... many of us learn
about the world from
social media.

interesting, personal,
informal, narrative-driven
yay :)



```
tweet_id.append(tweet_id)
tweet_time.append(tweet_time)
tweet_text.append(tweet_text)
tweet_coordinates.append(tweet_coordinates)
tweet_place.append(tweet_place)
tweet_count += 1

else:
    pass

et_dataframe['id'] = tweet_id
et_dataframe['time'] = tweet_time
et_dataframe['text'] = tweet_text
et_dataframe['coordinates'] = tweet_coordinates
et_dataframe['place'] = tweet_place

count)
dataframe

'road closed'
'locked',
'blocked road'
```



Tweetori (n.)

an emerging format for
explaining complex
concepts on Twitter

Hook

→
&it's hard



Tony Breu
@tony_breu · Follow

1/

"Why do my fingers get all wrinkly when I take a bath?" This was the question my 3-year-old daughter asked me recently after bath time.



Thread



Tony Breu @tony_breu · Jul 11, 2018

2/

Let me throw a version of my daughter's question out to you:

Which of the following contributes to "water immersion wrinkling"?

Water moves out of hand.

lets build an AI writing tool then.....
LLMs help people brainstorm & solve fixation...

so what features should the tool have???
let's do formative study (talk to users, read
literature, do annotation studies)....



5.8K



Reply



Copy link

Read 91 replies



Tony Breu @tony_breu · Jul 11, 2018

5/

It's true that water enters fingers after immersion. The portals of entry are likely sweat glands (which are highly concentrated in the hands).

Tweetorial Collection

TechTweets has hand-picked a collection of tweetorials on topics ranging from biology to linguistics that are useful for learning how to write tweetorials.

We've also annotated some exemplary tweetorials to shed light on what makes some of the

The Hook
Short and sweet, the first tweet "hooks" the reader with a personal story and intriguing question.



Jennifer Harrison?
@GeneticJen

So today I was talking about evolution being amazing and literally every animal being awesome and I'm told the dung beetle isn't. Thread 🧵

California Wildfires

— Written by @meehancrist on Nov 21, 2019. View on [Twitter](#).

hook

The author is talking about a concrete, relatable subject: the California wildfires of 2019, which is made more real with the accompanying image. But

she also makes it personal and emotional (it's her childhood home) and adds dramatic tension with "not in

the ways you expected." These techniques provide a reason to write

now, a clear subject for the tweetorial,

and an engaging question which makes the reader want to learn more.

Meehan Crist @meehancrist · Nov 21

I've been watching the landscape of my childhood burn with an aching heart, and wondering how much climate change is to blame. Turns out human activity is a major driver of California's wildfires, but not just in the ways you might imagine. (pic: Noah Berger) 1/10 <https://t.co/aX1ClJSrGi>



Tony Breu

@tony_breu · [Follow](#)

1/

"Why do my fingers get all wrinkly when I take a bath?" This was the question my 3-year-old daughter asked me recently after bath time.

I thought for a minute, then realized I didn't have a clue.

The explanation is so much cooler than I had expected...



[Read the full conversation on Twitter](#)

♥ 5.8K 💬 Reply 🔗 Copy link

[Read 91 replies](#)



no jargons!

visuals?

stories?

examples?

anecdotes?

suspense?

unanswered question?

Meehan Crist

@meehancrist · [Follow](#)

I've been watching the landscape of my childhood burn with an aching heart, and wondering how much climate change is to blame. Turns out human activity is a major driver of California's wildfires, but not just in the ways you might imagine. (pic: Noah Berger)

1/10



9:51 AM · Nov 21, 2019



[Read the full conversation on Twitter](#)

♥ 18 💬 Reply 🔗 Copy link

[Read 2 replies](#)



3 Requirements for an engaging hook:

➤ **R1:** Jargon-free

no jargons!



➤ **R2:** Relatable & Specific Example

visuals?

stories?

examples?

anecdotes?



➤ **R3:** Sparks Curiosity

suspense?

unanswered question?



- Step 1: Generate Topic
- Step 2: Generate Experiences
- Step 3: User Scenarios
- Step 4: Generate Anecdotes
- Step 5: Common Experiences
- Step 6: Generate Hooks
- Step 7: Write a Final Hook



Tweertorial Hook Incubator

Step 1: Input Topic & Generate Everyday Examples

[tweet_topic]:

language-model:

START

Here is a Quick Hook Generation:

Give me some ideas for your products and your brand to type next? That's what I'm looking for! I want to see how you can use your brand to create a hook that's not just a generic one, but one that's specific to your brand. I want to see how you can use your brand to create a hook that's not just a generic one, but one that's specific to your brand. I want to see how you can use your brand to create a hook that's not just a generic one, but one that's specific to your brand.

Here are 5 everyday examples:

1. Add something to your list of things to do.
2. Add something to your list of things to do.
3. Add something to your list of things to do.
4. Add something to your list of things to do.
5. Add something to your list of things to do.

REGENERATE

Notes (I save ideas)

What's the best way to generate a hook? I want to see how you can use your brand to create a hook that's not just a generic one, but one that's specific to your brand. I want to see how you can use your brand to create a hook that's not just a generic one, but one that's specific to your brand. I want to see how you can use your brand to create a hook that's not just a generic one, but one that's specific to your brand.

Step 2: Generate Common Experiences

[everyday_example]:

prompt:

language-model:

GENERATE COMMON EXPERIENCES

Here are 5 common experiences:

1. Unusually funny or unusual suggestions.
2. Unusually funny or unusual suggestions.
3. Unusually funny or unusual suggestions.
4. Unusually funny or unusual suggestions.
5. Unusually funny or unusual suggestions.

REGENERATE

Step 3: Generate User Scenarios

[common_experience]:

prompt:

language-model:

GENERATE USER SCENARIOS

Here are 5 user scenarios:

1. While writing an email, I suggested a word that "like a chicken".
2. A suggested "best morning with my cat and a cat".
3. A suggested a sentence "What you may not" with "you the my dog?"
4. Using to be a single girl, I added "what you the my dog?"
5. I suggested "best morning with my cat and a cat".
6. While typing "best morning with my cat and a cat".
7. While typing a comment when I suggested "I'd like a cat and a cat".
8. About about the weather, I suggested "I'd like a cat and a cat".
9. Documenting meeting notes, I suggested "I'd like a cat and a cat".
10. I suggested a sentence "What you may not" with "you the my dog?".
11. Writing a tweet about a cat, I suggested "I'd like a cat and a cat".
12. About for the purpose of "best morning with my cat and a cat".

REGENERATE

Step 4: Generate Personal Anecdotes

[user_scenario]:

prompt:

language-model:

GENERATE PERSONAL ANECDOTES

Here are 3 personal anecdotes:

1. While writing a tweet about a cat, I suggested "I'd like a cat and a cat".
2. A suggested "best morning with my cat and a cat".
3. A suggested "best morning with my cat and a cat".

REGENERATE

Step 5: Generate More Specific Anecdotes

[personal_anecdote]:

prompt:

language-model:

GENERATE SPECIFIC ANECDOTES

Here are 3 specific anecdotes:

1. While writing a tweet about a cat, I suggested "I'd like a cat and a cat".
2. A suggested "best morning with my cat and a cat".
3. A suggested "best morning with my cat and a cat".

REGENERATE

Step 6: Generate Example Hooks

[specific_anecdote]:

prompt:

language-model:

GENERATE EXAMPLE HOOKS

Here are 3 example hooks generated by GPT:

1. One tweet "best morning with my cat and a cat".
2. A suggested "best morning with my cat and a cat".
3. A suggested "best morning with my cat and a cat".

REGENERATE

Step 7: Write a Final Hook

[example_hook]:

prompt:

language-model:

GENERATE FINAL HOOK

Step 1: Input Topic to Generate Everyday Examples

Step 2: Generate Common Experiences

Step 3: Generate User Scenarios

Step 4: Generate Personal Anecdotes

Step 5: Generate More Specific Anecdotes

Step 6: Generate Example Hooks

Step 7: Write a Final Hook



When I fell ill just before my thesis deadline, drafting an email to ask for an extension was....



[tweet_topic]:	prompt:
AI	What are the ethical implications of artificial intelligence?
AI	How can we ensure the responsible use of AI technologies?
AI	What are the potential risks of AI in the workplace?
AI	How can AI be used to improve healthcare outcomes?
AI	What are the challenges of AI in education?
AI	How can AI be used to combat climate change?
AI	What are the ethical considerations of AI in law enforcement?
AI	How can AI be used to improve financial services?
AI	What are the potential benefits of AI in transportation?
AI	How can AI be used to improve social media experiences?
AI	What are the challenges of AI in cybersecurity?
AI	How can AI be used to improve customer service?
AI	What are the ethical implications of AI in journalism?
AI	How can AI be used to improve mental health services?
AI	What are the potential risks of AI in the military?
AI	How can AI be used to improve environmental monitoring?
AI	What are the challenges of AI in space exploration?
AI	How can AI be used to improve disaster response?
AI	What are the ethical considerations of AI in sports?
AI	How can AI be used to improve agricultural productivity?
AI	What are the potential benefits of AI in energy production?
AI	How can AI be used to improve infrastructure management?
AI	What are the challenges of AI in urban planning?
AI	How can AI be used to improve public safety?
AI	What are the ethical implications of AI in the arts?
AI	How can AI be used to improve language learning?
AI	What are the potential risks of AI in the legal system?
AI	How can AI be used to improve historical research?
AI	What are the challenges of AI in the entertainment industry?
AI	How can AI be used to improve scientific discovery?
AI	What are the potential benefits of AI in the space industry?
AI	How can AI be used to improve the quality of life in aging populations?
AI	What are the ethical considerations of AI in the automotive industry?
AI	How can AI be used to improve the efficiency of supply chains?
AI	What are the challenges of AI in the telecommunications industry?
AI	How can AI be used to improve the accuracy of weather forecasting?
AI	What are the potential risks of AI in the financial industry?
AI	How can AI be used to improve the quality of education in developing countries?
AI	What are the ethical implications of AI in the healthcare industry?
AI	How can AI be used to improve the efficiency of government services?
AI	What are the challenges of AI in the energy industry?
AI	How can AI be used to improve the quality of life in rural communities?
AI	What are the potential benefits of AI in the manufacturing industry?
AI	How can AI be used to improve the accuracy of social media algorithms?
AI	What are the challenges of AI in the aerospace industry?
AI	How can AI be used to improve the quality of life in urban areas?
AI	What are the ethical considerations of AI in the media industry?
AI	How can AI be used to improve the efficiency of logistics operations?
AI	What are the potential risks of AI in the defense industry?
AI	How can AI be used to improve the quality of life in the elderly population?
AI	What are the challenges of AI in the construction industry?
AI	How can AI be used to improve the accuracy of financial forecasting?
AI	What are the potential benefits of AI in the food industry?
AI	How can AI be used to improve the quality of life in the digital age?
AI	What are the ethical implications of AI in the biotechnology industry?
AI	How can AI be used to improve the efficiency of the transportation system?
AI	What are the challenges of AI in the information technology industry?
AI	How can AI be used to improve the quality of life in the global community?
AI	What are the potential risks of AI in the future of work?
AI	How can AI be used to improve the accuracy of the legal system?
AI	What are the challenges of AI in the space exploration industry?
AI	How can AI be used to improve the quality of life in the 21st century?
AI	What are the ethical considerations of AI in the future of humanity?
AI	How can AI be used to improve the efficiency of the global economy?
AI	What are the potential benefits of AI in the future of the planet?
AI	How can AI be used to improve the quality of life in the future generations?
AI	What are the challenges of AI in the future of the world?
AI	How can AI be used to improve the accuracy of the future of the universe?
AI	What are the potential risks of AI in the future of the cosmos?
AI	How can AI be used to improve the quality of life in the future of the universe?
AI	What are the ethical implications of AI in the future of the universe?
AI	How can AI be used to improve the efficiency of the future of the universe?
AI	What are the challenges of AI in the future of the universe?
AI	How can AI be used to improve the quality of life in the future of the universe?
AI	What are the potential benefits of AI in the future of the universe?
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AI	What are the potential risks of AI in the future of the universe?
AI	How can AI be used to improve the accuracy of the future of the universe?
AI	What are the challenges of AI in the future of the universe?</

[tweet_topic]:	prompt:
language models	Please list 5 everyday examples of [tweet_topic]. Each example should be under 10 words, related to a general audience, and not require technical knowledge.
START	

Here is a Quick Book Generation:

Ever wonder how your phone predicts what you're about to type next? 🤖 That's thanks to language models! Think of it as your device's mini fortune-teller, guessing your next word based on what you've typed so far. But how can it predict your personal words? Look out for our upcoming blog to dive into the mystery! 🕵️
 @knowwhatyouknow @k10 @lucachanTech

Here are 5 everyday examples:

1. Auto-correcting typos on your smartphone
2. Predictive texting while writing messages
3. Voice recognition like Siri or Alexa
4. Search engine suggestions
5. Email spam filters

MIGRATE

Step 2: Generate Common Experiences

leveraging examples

The diagram illustrates the LLM workflow for generating common experiences. It consists of two main components: a **prompt** and a **response**.

prompt:

```
[everyday_example]:
```

Please list 5 common experiences with using [everyday_example]. Each experience should be under 10 words, related to a general software, and not require technical knowledge.

response:

Predictive texts and just suggestions for creative writing

GENERATE COMMON EXPERIENCES

Here are 5 common experiences:

1	Unexpectedly humorous or unusual suggestions.	2 Capped to Head Stop
2	Autocorrect mistakes changing intended meaning.	♡
3	Helpful in faster typing and spelling correction.	♡
4	Stunting creative expression with standard phrases.	♡
5	Learning to use unusual words or phrases less often.	♡

618.78

Step 3: Generate User Scenarios

[common_experience]:	prompt:
Unexpectedly humorous or nonsensical suggestions	Please list 12 user interaction scenarios, encounters with [common_experience] while using [tool_topic]. Each scenario should be under 15 words, from a first-person perspective, related to a general audience, and not require technical knowledge.
GENERATE USER SCENARIOS	

Here are 5 user scenarios:

- 1 While writing an email, I suggested (and used) the phrase "a dinosaur". 
- 2 I responded "Sweet morning" with "Party with someone". 
- 3 I suggested to someone "Will you marry me?" with "Can you be my partner?" 
- 4 Using it as a movie quote, I said "I have come here for a minute in coming days". 
- 5 I responded "I have been with you" with "I have been with you". 
- 6 While typing "I have been with you" I completed it as "I have been with you". 
- 7 While typing a comment about it, I suggested "Let's have a party party". 
- 8 I responded to the question "What is your name?" with "Cloudy the character of 'Fairy Tales'". 
- 9 Documenting meeting minutes, I mentioned "Meeting was an interesting conversation". 
- 10 I suggested using "Sweet morning" as a headline for "City of Engineers" in my newspaper. 
- 11 Writing a heartfelt poem, I ended with "Let us be the friends who are". 
- 12 Asked for the phrase of "happy", I suggested "Sweet 'Fairy Tales' conversation". 

Step 4: Generate Personal Anecdotes

[user.scrnami]:	prompt:
When typing "Guardians of", it completed as "Guardians of the Galaxy" by Ben.	Please write 3 short personal anecdotes about the topic, [topic_tag], for the user scenario of [user_scenario] for creative writing. Each anecdote should be under 30 words, from a creative writer's perspective, related to a general audience, and not require technical knowledge. Make sure to mention that the writer actually incorporated the funny suggestion by their future plot.
GENERATE PERSONAL ANECDOTES	

Here are the 3 personal anecdotes:

1. While crafting a superhero novel, I typed "Guardians of," intending to type "Guardians of the Galaxy," but my language model amusingly suggested "Guardians of the Gummi Bears" (it spelled as close for a humorous subject).
2. I was creating a mythical saga and passed at "Guardians of," hilariously, the language model completed it as "Guardians of the Gummi Bears" I worked it in as a fantastical moment.
3. Absurdly, adding "Guardians of," into a political satire, the language model proposed "Guardians of the Gummi Bears." I laughed out loud and turned it into a whimsical political parody.

Step 5: Generate More Specific Anecdotes

[personal_anecdote]:	prompt:
While crafting a superhero novel, I typed "Guardians of..." intending to type "Guardians of the Galaxy," but my language model amusingly suggested "Guardians of the Curry Feast." It sparked an idea for a humorous subplot.	Finest rewrite 3 versions of this anecdote to make it more specific. Replace all generic terms with concrete and specific instances: [personal_anecdote].

Here are 3 specific versions:

1 While I was diligently typing my letter before me, my fingers swiftly moved to type the famous phrase "Queens of the Galaxy" which I clearly adored and was extremely impressed upon. "Queens of the Galaxy Machine". This sentence appeared in printing across for a while waiting for me to move on.

2 Once during an intense session of composing my letter, I started to type "I agree with you" and then "Queens of the Galaxy" which I clearly adored and was extremely impressed upon. With my computer, with its 60 language model, I started to type "Queens of the Galaxy" and then "Queens of the Galaxy" which I clearly adored and was extremely impressed upon. I started to type "Queens of the Galaxy" which I clearly adored and was extremely impressed upon. I started to type "Queens of the Galaxy" which I clearly adored and was extremely impressed upon.

3 One day, when I was passionately typing an English letter, I was all set to reference the quote "Queens of the Galaxy" which I clearly adored and was extremely impressed upon. I started to type "Queens of the Galaxy" which I clearly adored and was extremely impressed upon. I started to type "Queens of the Galaxy" which I clearly adored and was extremely impressed upon. I started to type "Queens of the Galaxy" which I clearly adored and was extremely impressed upon.

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Step 6: Generate Example Hooks

specific anecdote(s):	prompt:
There was, when crafting a social media "I signed a contract" of "relating to the fact 'Standards of the Last'," that my language module amusingly generated "Standards of the Tummy Rinsing" I spotted an idea for a homework subject. Why? WHY NOT MAKE IT SPREAD AND WITH A PIGGLE. There's a whole chapter about the gummy bear language 熊の舌	Here is a personal anecdote about the topic <code>love, topic</code>, for the experience of <code>(common_experience)</code>: <code>[specific_anecdote]</code>. Based on this anecdote, please write 2 versions of the opening tweet to introduce the excerpt of <code>Love_Topic</code>. The tweet should have as <code>[topic]</code> to attract attention a specific and reliable example. It should have a driving question to spark curiosity and make readers want to read more. Make sure to include a few emojis and all the details.

Here are 3 example books generated by GPT:

[illegible]

Step 7: Write a Final Hook

Ever typed "Guardians of" and expected "the Galaxy" but got "the Gummi Bears" suggested? 🤖👾 Just got sugary space heroes featured in my new sci-fi SO CUTE! or what if I told you, that was no random suggestion but a brilliant technology called "Language Models" 🤖👾

100%

Input to the next steps

Step output

[tweet_topic]
"language models"

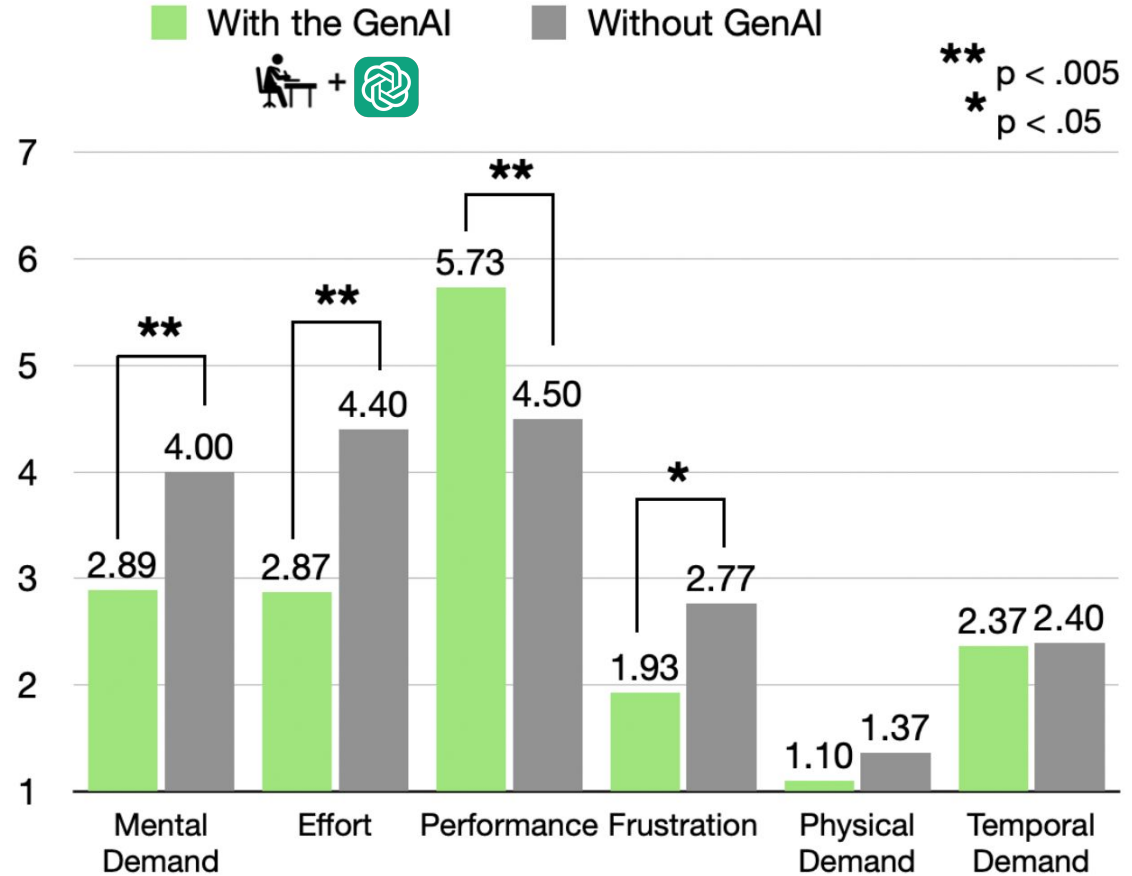
Step 1: Input Topic to Generate Everyday Examples

[everyday_example] ● —————
"autocomplete feature in gmail"



Step 2: Generate Common Experiences

- ↓ Reduced mental demand
- ↓ Reduced effort
- ↓ Reduced frustration
- ↑ Improved performance



1. find a user problem

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2. understand why the problem is hard to solve
 - formative interview, user research, field/observation study
3. understand how to solve it
4. design/develop a tool
 - prototyping, testing, iterating (with users sometimes)
 - deployment
5. test the tool
 - usability (user) testing
 - annotation study, ablation study, etc

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 - annotation study, ablation study, etc
 - build a testable UI/app/webapp
 - recruit users & run testing

Deployment.

[illegible]

How to find users? How to do user testing?

- **Targeted audience group.**
 - sampling... science.....
 - IRB ethics approval

Between-Subjects

vs.

Within-Subjects

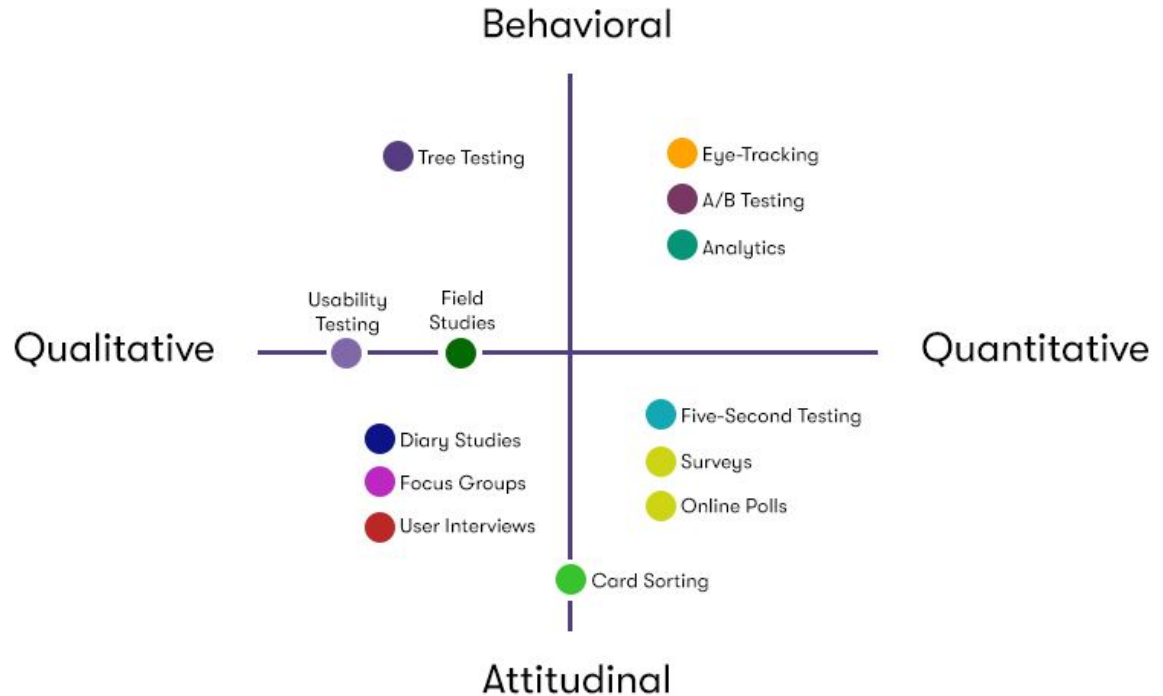


Site 1

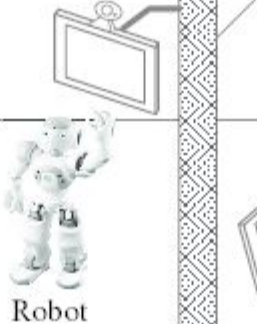


Diary Study: Timeline of Activities

UX Research Methods



Experiment Room



Robot

How to find users? How to do user testing?

- **Targeted audience group.**
 - sampling... science.....
 - IRB ethics approval
- **User study**
 - Introduction to the targeted problem case, the tool, use case,
 - Demo
 - Usage session
 - Methods:
 - **Data collection**
 - Usage data (log, time spent etc)
 - Task experience (how stressful, cognitive load, etc)
 - System experience (usefulness, ease of usage, performance, etc)

tyty

Sruthi

- 3rd year CS PhD working on **CV/Robotics**
- Advised by: Carl Vondrick & Richard Zemel
- Research interests: Learning ML models that are generalizable
- Undergrad @ Georgia Tech '22
 - Computer science
- Life interests:
 - Dance (indian classical)
 - Volunteering

Outline

1. Past research work
2. Current research work
3. How can you use AI?

Self-Learning Robots

Images



DALL-E

Videos



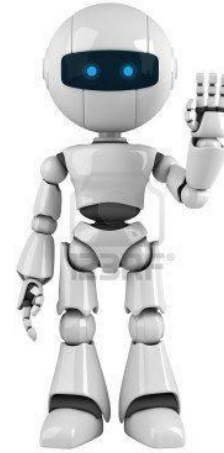
SORA

Language



ChatGPT

Robots?



Current research work

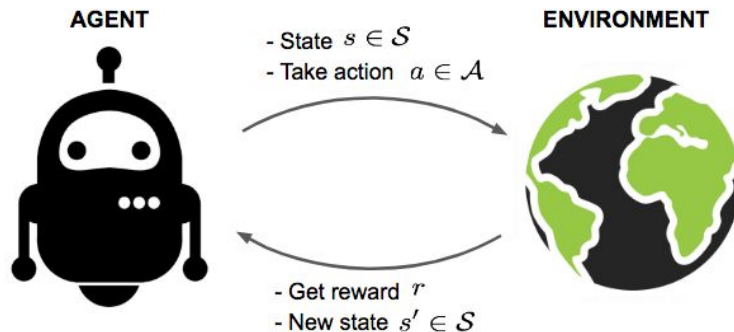
Why are robots hard?

1. Not enough training data
2. Many different morphologies
3. Has to *actually* work

What works so far?

1. Reinforcement Learning

- a. Good rewards hard to design
- b. Rewards may be sparse
- c. Hard to train (esp in the earlier stages)



Current research work

What works so far?

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monkey see

2. Behavior Cloning

- a. Much stronger starting point
- b. Distributional mismatch leads to compounding errors
- c. Humans may not be the experts



monkey do

Behavior Cloning + Self-Learning

Given expert demonstrations and a trained policy on task A, how does the agent improve through self-directed learning?

Behavior Cloning on task A, self-adapt to task B



Current research work

Behavior Cloning + Self-Learning

Given expert demonstrations and a trained policy on task A, how does the agent improve through self-directed learning?

1. Learn from failures?
2. Do better than an expert task?

Method

Procedure:

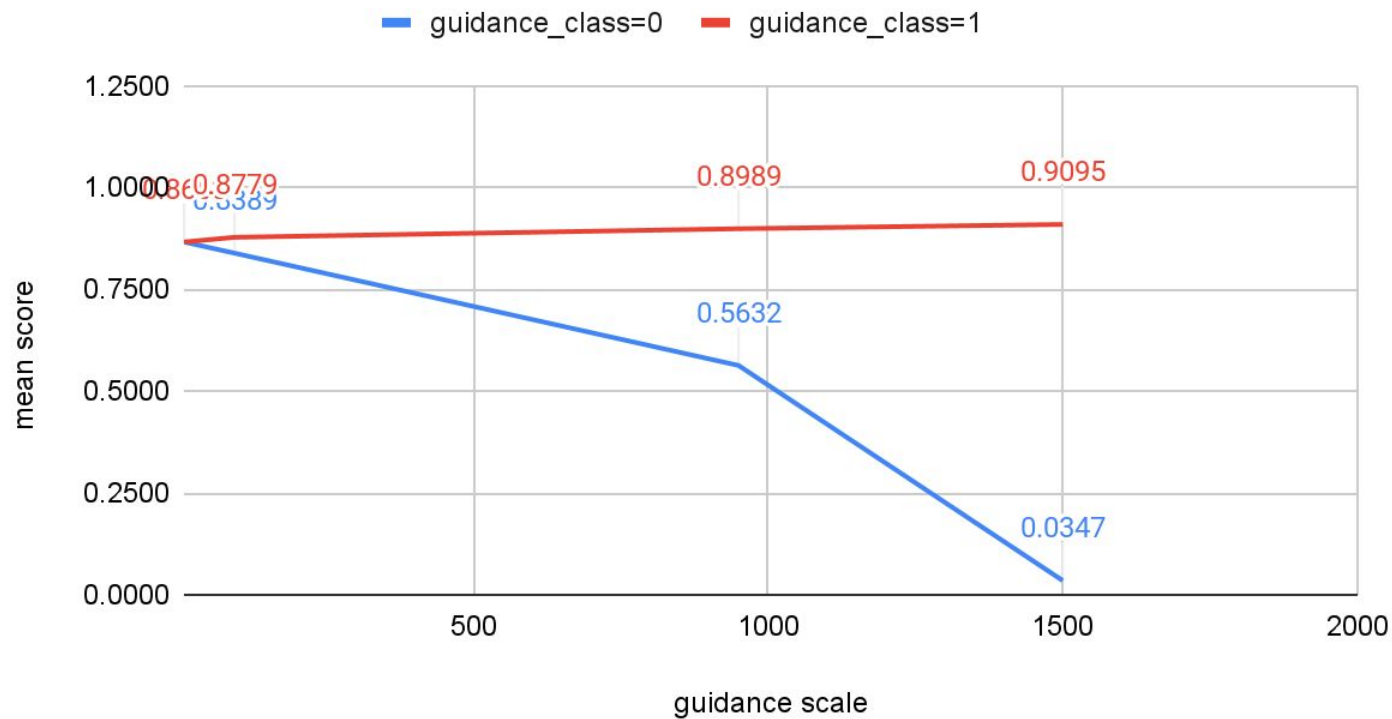
1. Initialize policy to some current best policy
2. Sample current best policy to collect a dataset
3. Update policy
4. Repeat 2-4

Method

Procedure:

1. Initialize policy to some current best policy
2. **Sample** current best policy to collect a dataset
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mean score vs. guidance scale



Method

Procedure:

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4. Repeat 2-4

How can you use AI?

1. Use HuggingFace demos as a service
2. Run a model as a service
3. Using existing frameworks to develop applications with latest research

Why RAG?

1. Most products that we use could benefit from chatbots
2. LLMs are not grounded in facts/have outdated information
3. May not have information relevant to your company

Ex: Lecturer on a philosophy of thought

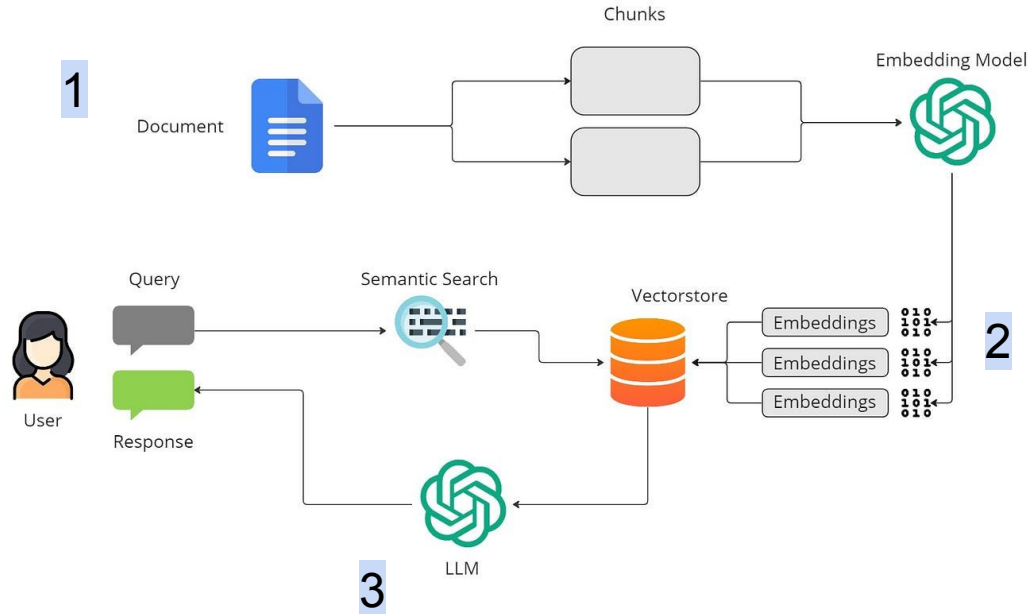
How can you use AI?

What is RAG?

1. Given: Database of documents with relevant information
2. Want: Chatbot that can answer questions based on this knowledge

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How can you use AI?

How does RAG work?

1. Indexing & Chunking:
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4. Prompting:
 - a. Design a prompt including user query + relevant data, and query LLM
5. Ex:

```
template = """Answer the question based only on  
the following context: {context}  
Question: {question}  
"""
```

How can you use AI?

LangChain